Python: If, Elif and Else

Teach your program to make decisions.

Python | Beginner | Ages 9–16

InfiBee Coders

The Big Idea

- A decision lets code do different things in different situations.
- `if` runs a block only when its condition is True.
- `elif` checks another condition. `else` runs when nothing above matched.
- The lines inside are indented by 4 spaces. A colon `:` ends each if, elif and else line.
- Python checks from the top and stops at the first True condition.

Comparison Operators

- `==` equal to, `!=` not equal to.
- `>` greater than, `<` less than.
- `>=` greater than or equal to, `<=` less than or equal to.
- Remember: one `=` stores a value, two `==` compares.

Combining Conditions

- `and` is True only if both sides are True.
- `or` is True if at least one side is True.
- `not` flips True to False and False to True.
- `in` checks membership: `"a" in "cat"` is True.
- Brackets make complex conditions easier to read.

The Shape of an If Statement

- `if condition:` then an indented block.
- `elif other_condition:` is optional and can repeat.
- `else:` is optional and goes last.
- Blocks can sit inside blocks. This is called nesting.

💻 Code Examples

A simple if / else — python

```python
age = 13
if age >= 13:
    print("You can join!")
else:
    print("Sorry, too young.")
```

▶ **Output:**

```
You can join!
```

13 >= 13 is True, so the first block runs.

elif for several choices — python

```python
score = 82
if score >= 90:
    grade = "A"
elif score >= 75:
    grade = "B"
elif score >= 50:
    grade = "C"
else:
    grade = "Try again"
print("Grade:", grade)
```

▶ **Output:**

```
Grade: B
```

Python stops at the first condition that is True.

Even or odd with % — python

```python
for n in [4, 7]:
    if n % 2 == 0:
        print(n, "is even")
    else:
        print(n, "is odd")
```

▶ **Output:**

```
4 is even
7 is odd
```

n % 2 is the remainder when you divide by 2.

and, or, not python

```python
age = 11
has_ticket = True
if age >= 10 and has_ticket:
    print("Enter")
if not has_ticket or age < 5:
    print("Stay outside")
print("Done")
```

▶ **Output:**

```
Enter
Done
```

Only the first message prints.

Checking user input python

```python
password = input("Password: ")
if password == "python123":
    print("Welcome back!")
else:
    print("Wrong password")
```

▶ **Output:**

```
Password: python123
Welcome back!
```

Try the correct password and a wrong one.

Nested if python

```python
weather = "sunny"
temp = 31
if weather == "sunny":
    if temp > 30:
        print("Wear a hat")
    else:
        print("Nice day")
```

▶ **Output:**

```
Wear a hat
```

The second if only runs when the first is True.

📚 Key Terms

Condition
A question whose answer is True or False.

Boolean
A value that is True or False.

Comparison
Checking how two values relate.

and / or / not
Logic words that combine conditions.

Indentation
The spaces that show what is inside a block.

Nested
One block inside another.

elif
Short for else if.

Branch
One possible path through the code.

else
What runs when nothing above matched.

Equality
Checking if two values are the same with ==.

🧮 Quick Reference

```
if x > 5:
    print("big")
elif x == 5:
    print("five")
else:
    print("small")
```

```
a and b   a or b   not a
x == y   x != y   x >= y
```

✏️ Worked Examples

Q: Write a program that says whether a number is positive, negative or zero.

A: Use if n > 0, elif n < 0, else. The else covers zero.

Q: Decide ticket prices: under 5 free, 5 to 12 half price, otherwise full.

A: if age < 5, elif age <= 12, else. The elif only runs when age is 5 or more.

Q: Why does if x = 5: give an error?

A: A single = stores a value. Comparisons need ==, so use if x == 5:.

🚫 Common Mistakes

- Using `=` instead of `==` in a condition.
- Forgetting the colon at the end of the if, elif or else line.
- Wrong indentation, which causes an IndentationError.
- Writing `if x == 1 or 2:`. Write `if x == 1 or x == 2:` instead.
- Putting `else` before `elif`.

👨‍👩‍👧 Parent Corner

- Ask What if? questions in daily life, like if it rains, then we take an umbrella, else we walk.
- Ask your child to write a program that decides what to wear based on the temperature.
- Practise reading code aloud: if age is at least 13, then print welcome.
- Try a number guessing game together using if, elif and else.
- Praise precise thinking, such as spotting an edge case.

🎯 Quick Check

1. Which operator checks if two values are equal?

✅ == (two equal signs).

2. What does elif mean?

✅ else if: another condition to check.

3. What ends an if line?

✅ A colon.

4. What does True and False give?

✅ False. The word and needs both sides to be True.

5. What runs when no condition matched?

✅ The else block.